

MTH302 Business Mathematics & Statistics

All formulas

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Formula 1

Percent = fraction x 100

Formula 2

Percent = base x rate

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Percent = fraction x 100

Formula 2

Percent = base x rate

Formula 3

Average = sum / number Where Sum = sum of all data values Number = number of all data values

Formula 4

Change = final value - initial value % change = change x 100 / initial value Or % change = (final value - initial value) x 100 / initial value

Formula 5

Stock yield = annual dividend payments / stock's current share price

Formula 6

Earnings per share = total profits of company / number of shares

Formula 7

Price earnings ratio = market value per share (or) company's current share price / Earnings per share Or Price earnings ratio = market value per share (or) company's current share price / total profits of company / number of shares Or Price earning ratio = market value per share (or) company's current share price x number of shares / total profits of company

Formula 8

Net current asset value per share = (current assets – total liabilities) / number of shares outstanding

Formula 9

Dividends = Dividends% x number of shares / face value of the share

Formula 10

Return on investment in % = total gain x 100 / total cost

Formula 11

Net cost price = list price – discount in Rs. Where discount in Rs = discount % x list price So Net cost price = list price - (discount % x list price)

Formula 12

Simple interest = principal x time in years x rate of interest per annum / 100 Or $I = \frac{PRT}{100}$

Formula 13

Compound interest = S-P Where $S = P (1 + R/100)^n$ So Compound interest = $P (1 + R/100)^n - P$
 S = money accrued after n years or compound amount or accumulated value

P = principal R = rate of interest per annum N = number of periods

Formula 14

$PV_{\text{ordinary annuity (OR) } A = r \text{ (OR) } C \times \text{DISCOUNT FACTOR}}$

$\text{DISCOUNT FACTOR} = [1 - (1+i)^{-n}] / i$ $PV_{\text{ordinary annuity (OR) } A = r \text{ (OR) } C \times [1 - (1+i)^{-n}] / i$ A = discounted or present worth of an annuity

Formula 15

$FV_{\text{ordinary annuity (OR) } A = R \text{ (OR) } C \times \text{ACCUMULATION FACTOR}}$ Accumulation factor = $[(1+i)^n - 1] / i$
 $FV_{\text{ordinary annuity (OR) } S = r \text{ (OR) } C \times [(1+i)^n - 1] / i$ S = accumulated value C (OR) r = payments per

periods (OR) amount of annuity (OR) cash flow per period i = rate of interest per annum n = number of payments

Formula 16

Selling price = cost price + Rs. Markup on cost
 Rs. Markup on cost = cost price x % markup on cost
 So Selling price = cost price + (cost price x % markup on cost)
 So Selling price = cost price (1 + % markup on cost)

Formula 17

Selling price = cost price + (selling price x % markup on sale)

cost price = Selling price - (Selling price x % markup on sale)
 Cost price = Selling price (1 - % markup on sale)

Formula 18

Rs. Markup = Selling price - Cost price (OR) Rs. Markup on cost = cost price x % markup on cost
 Rs. Markup on sale = Selling price x % markup on sale

Formula 19

% Markup on cost = (Selling price - Cost price) x 100 / Cost price (OR) Rs. Markup on cost = Rs. Markup x 100 / Cost price (AND) % Markup on SALE = (Selling price - Cost price) x 100 / Selling price (OR) Rs. Markup on sale = Rs. Markup x 100 / Selling price

Formula 20

New selling price = current (OR) old selling price - Rs. Markdown
 Where Rs. Markdown = % Markdown x current (OR) old selling price
 New selling price = current (OR) old selling price - (% Markdown x current (OR) old selling price)
 New selling price = current (OR) old selling price (1 - % Markdown)

Formula 21

Rs. Markdown = current (OR) old (OR) original selling price - new selling price
 % Markdown = Rs. Markdown x 100 / current (OR) old (OR) original selling price

Formula 22

Actual Rs. Paid = total Rs. Assumed to be paid due to discount (1 - % discount)

Formula 23

Margin % = Rs. markup (OR) Rs. Margin x 100 / sale
 And Rs. Margin = Margin % x selling price
 While markup % = Rs. Markup x 100 / cost
 Margin (OR) markup = (Selling price - Cost price) x 100 / Selling price
 Selling price = cost price + Rs. Margin / Rs. Markup
Remember unless it is mentioned that markup is on sale, simple markup means markup on cost while margin is always on sale

Formula 24

Markup on sale = $\% \text{ markup on cost} / (1 + \% \text{ markup on cost})$
 $\% \text{ markup on cost} = \% \text{ markup on sale} / (1 + \% \text{ markup on sale})$

Formula 25

Break even point (OR) BEP in units = $\text{fixed cost} / \text{contribution margin per unit}$

Formula 26

BEP in Rs. = $\text{fixed cost} \times \text{net sales} / \text{TOTAL contribution margin}$
 BEP in Rs. = $\text{fixed cost} \times \text{selling price per unit} / \text{contribution margin per unit}$

Formula 27

BEP as $\% \text{ capacity} = \text{BEP in units} \times 100 / \text{production capacity}$

Formula 28

Total Contribution margin = $\text{Net sales} - \text{variable cost}$
 Contribution margin per unit = $\text{selling price per unit} - \text{variable cost per unit}$

Formula 29

Contribution rate = $\text{Total Contribution margin} \times 100 / \text{net sales}$
 Contribution rate = $\text{Contribution margin per unit} \times 100 / \text{selling price per unit}$

Formula 30

Net income = $\text{number of units sold above BEP} \times \text{Contribution margin per unit}$

Formula 31

Net loss = $\text{number of units sold below BEP} \times \text{Contribution margin per unit}$
 Net loss = $-\text{Net income} = -\text{number of units sold above BEP} \times \text{Contribution margin per unit}$
 Average = $\text{sum} / \text{number}$ Where Sum = $\text{sum of all data values}$ Number = $\text{number of all data values}$

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Change = $\text{final value} - \text{initial value}$ $\% \text{ change} = \text{change} \times 100 / \text{initial value}$ Or $\% \text{ change} = (\text{final value} - \text{initial value}) \times 100 / \text{initial value}$

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BEP as % capacity = BEP in units x 100 / production capacity

Formula 28

Total Contribution margin = Net sales - variable cost

Contribution margin per unit = selling price per unit - variable cost per unit

Formula 29

Contribution rate = Total Contribution margin x 100 / net sales

Contribution rate = Contribution margin per unit x 100 / selling price per unit

Formula 30

Net income = number of units sold above BEP x Contribution margin per unit

Formula 31

Net loss = number of units sold below BEP x Contribution margin per unit

Net loss = - Net income = - number of units sold above BEP x Contribution margin per unit